THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 3: **Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Heating a sulphide ore in air is: 1

(a) roasting **(b)** refining **(c)** smelting **(d)** calcination

2. Which non-metal is a good conductor of electricity? 1

(a) phosphorus **(b)** sulphur **(c)** iodine **(d)** graphite

3. An alloy of mercury is called: 1

(a) solder **(b)** bronze **(c)** brass **(d)** amalgam

4. Heating a carbonate ore in limited air is: 1

(a) refining **(b)** calcination **(c)** roasting **(d)** electrolysis

5. Which is the hardest natural substance? 1

(a) gold **(b)** iron **(c)** graphite **(d)** diamond

6. Brass contains copper and: 1

(a) lead **(b)** nickel **(c)** tin **(d)** zinc

7. Pure metal is obtained from impure by: 1

(a) galvanisation **(b)** alloying **(c)** refining **(d)** roasting

8. Which metal is the softest? 1

(a) copper **(b)** iron **(c)** gold **(d)** sodium

9. The metal used for galvanisation is: 1

(a) tin **(b)** copper **(c)** lead **(d)** zinc

10. Reaction of metal with oxygen gives metal: 1

(a) oxide **(b)** chloride **(c)** sulphide **(d)** hydroxide

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define roasting and calcination. 2

12. What is an alloy? Give one example. 2

13. Name two metals that react with cold water. 2

14. What is meant by galvanisation? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the equation for the reaction of zinc with dilute HCl and name the gas. 3

16. Define: 3
(i) mineral
(ii) ore
(iii) gangue.

17. Which gas is produced when metals react with dilute acids? Give a test. 3

OR

Write the equation for the reaction of iron with copper sulphate solution and give the colour change. 3

18. State three ways to prevent rusting of iron. 3

OR

What are ionic compounds? Give two examples. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain corrosion and rusting. Mention conditions and three ways of prevention. 5

OR

Explain what are acidic, basic, neutral and amphoteric oxides with one example each. 5

20. Write the steps in the extraction of metals from ore for a metal of medium reactivity such as zinc. 5

OR

Describe the electrolytic refining of copper in simple terms: electrodes, electrolyte and result. 5

END OF QUESTION PAPER — Total: 40 Marks

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